# Keshav Singh

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## PROFILE

DevOps Engineer with 4+ years of professional experience in Java application development, deployment automation, cloud infrastructure and containerized applications. Hands-on experience with AWS, Terraform, Jenkins, Docker and Kubernetes, including deployment troubleshooting, monitoring and secure access controls. AWS Certified DevOps Engineer – Professional with an MSc in Cloud Computing and independent projects in AWS cost optimization and Kubernetes diagnostics.

## CORE SKILLS

**Cloud & Infrastructure:** AWS (EC2, S3, IAM, VPC, RDS, EKS, ECR, ALB), Terraform
**Azure & Networking:** Azure provisioning with Terraform, Azure Monitor, AWS VPC, load balancing
**CI/CD & Containers:** Jenkins, GitHub Actions, Docker, Kubernetes, Helm
**Monitoring & Operations:** Prometheus, Grafana, CloudWatch, Azure Monitor, Linux
**Programming & Automation:** Python, Bash, Java, TypeScript, SQL
**Development & Testing:** React, Vite, FastAPI, boto3, REST APIs, pytest, Git
**Security & Access:** AWS IAM, Kubernetes RBAC, secrets management, GitHub OIDC

## EXPERIENCE

**DevOps Engineer** Jan 2025 – Aug 2026
*Neosoft Systems & Cloud Services* *Remote*

- Managed Jenkins and GitHub Actions pipelines to automate application builds, tests and deployments across development, testing and production, reducing manual release effort by approximately 40%.
- Provisioned AWS infrastructure with Terraform across EC2, S3, IAM and RDS, making environment setup repeatable and reducing manual configuration.
- Maintained AWS VPC networking and load-balancing configurations to support application connectivity; investigated networking issues affecting cloud workloads.
- Supported Docker and Kubernetes applications, managing deployments, configuration, resource limits, health probes and rolling updates across production and non-production environments.
- Developed Python and Bash scripts for deployment validation, log collection, environment checks and routine operational tasks, saving approximately 8–10 hours of manual effort per month.
- Created CloudWatch, Prometheus and Grafana dashboards and alerts to monitor applications and infrastructure; correlated metrics and logs during incident investigation.
- Performed pre-deployment checks and post-release validation, investigated failed deployments and coordinated rollback procedures; documented release and recovery steps in runbooks.
- Applied AWS IAM, Kubernetes RBAC, secrets management and least-privilege permissions; investigated Linux, AWS and container incidents using system logs and Kubernetes diagnostics.

**Software Engineer** Nov 2020 – Mar 2023
*Zensar Technologies* *Pune, India*

- Started in Java application development and transitioned into DevOps responsibilities, moving from application maintenance and debugging to build automation, deployment and infrastructure support.
- Developed and maintained Java application code, investigated defects and tested changes before release, building an understanding of application behavior and release requirements.
- Designed and maintained 25+ CI/CD pipelines to automate build and deployment workflows, reducing deployment time from 45 minutes to under 10 minutes.
- Automated AWS and Azure environment provisioning with Terraform, replacing manual infrastructure setup with repeatable configuration and supporting consistent application environments.
- Supported containerized microservices on Kubernetes using Helm, autoscaling, health probes and controlled deployment strategies.

- Implemented IAM, RBAC, secrets management and vulnerability scanning as part of access and deployment security controls.
- Troubleshot application build, deployment and infrastructure incidents, using development experience to investigate failures across application and operational components.
- Built monitoring and alerting for 20+ services using Prometheus, Grafana, CloudWatch and Azure Monitor to improve incident detection and operational visibility.

## PROJECTS

**CloudSpend-Guard – AWS Cost Optimization** Aug 2026 – Present
*Independent Project – GitHub Repository* *AWS / Python / DevOps*

- Built a FastAPI platform using boto3 to identify low-utilization EC2 instances and unattached EBS volumes using CloudWatch metrics and AWS pricing data.
- Developed a React and TypeScript dashboard to display discovered resources, cost optimization recommendations and estimated savings.
- Provisioned AWS infrastructure with Terraform and deployed Docker workloads to private Amazon EKS using Helm, ECR and an ALB.
- Automated testing, image builds and ECR publishing with GitHub Actions, OIDC authentication, least-privilege IAM and immutable Git-SHA image tags.
- Tested resource discovery and recommendation APIs with pytest to validate backend behavior during development.

**KubeDiagnose – Kubernetes Diagnostics CLI**
*Independent Project – GitHub Repository* *Kubernetes / Troubleshooting*

- Built a Python CLI to investigate Kubernetes workload failures, with workload and namespace selection through the `diagnose` command.
- Created demonstration workloads for CrashLoopBackOff and image pull failures to exercise application startup and container image troubleshooting scenarios.
- Added scenarios for out-of-memory errors, failed health probes and scheduling issues to demonstrate resource, workload health and placement troubleshooting.
- Documented CLI usage and diagnostic examples with screenshots, making troubleshooting scenarios reproducible for users reviewing the project.

## CERTIFICATION

**AWS Certified DevOps Engineer – Professional**

## EDUCATION

**National College of Ireland** Dublin, Ireland
*MSc in Cloud Computing* 2023 – 2024